EQUIVALENCE OF UNBOUNDED DERIVED CATEGORIES OF SUITABLE SUBCATEGORIES, IN THE FIRST VARIANT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; both steps are inside the counted span and the situation cited by the statement is the one introduced earlier in the chapter.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The situation of the Olsson-Laszlo comparison: site, sheaf of rings and weak Serre subcategory with vanishing bounds. Same chapter, the situation introduced earlier in the chapter, which is cited by the counted statement and whose content is the vanishing bound hypothesis used in the proof. Supplies hypotheses (4) and (5) of the statement and the input of the companion results.
- Bounded version of the equivalence of derived categories of subcategories, and the recovery of objects as derived limits of their truncations. Same chapter, the companion bounded statement and the companion result recovering an object of the subcategory as the derived limit of its truncations. Supplies the two ingredients from which the unbounded statement is assembled.

Source reference. Original reference 0D6R: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 5687–5687. Navigation pointer invoked at proof line 6450; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0D7U: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 6391–6391. Navigation pointer invoked at proof

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line 6466; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0D6U: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 5796–5796. Navigation pointer invoked at proof line 6467; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $f : (Sh(\mathcal{C}), \mathcal{O}) \to (Sh(\mathcal{C}'), \mathcal{O}')$ be a morphism of ringed topoi. Let $\mathcal{A} \subset Mod(\mathcal{O})$ and $\mathcal{A}' \subset Mod(\mathcal{O}')$ be weak Serre subcategories. Assume*

(1) *f is flat,*
(2) *f^* induces an equivalence of categories $\mathcal{A}' \to \mathcal{A}$,*
(3) *$\mathcal{F}' \to Rf_*f^*\mathcal{F}'$ is an isomorphism for $\mathcal{F}' \in \mathrm{Ob}(\mathcal{A}')$,*
(4) *$\mathcal{C}, \mathcal{O}, \mathcal{A}$ satisfy the assumption of Situation 0D6R,*
(5) *$\mathcal{C}', \mathcal{O}', \mathcal{A}'$ satisfy the assumption of Situation 0D6R.*

Then $f^* : D_{\mathcal{A}'}(\mathcal{O}') \to D_{\mathcal{A}}(\mathcal{O})$ is an equivalence of categories with quasi-inverse given by $Rf_* : D_{\mathcal{A}}(\mathcal{O}) \to D_{\mathcal{A}'}(\mathcal{O}')$.

Proof. Since f^* is exact, it is clear that f^* defines a functor $f^* : D_{\mathcal{A}'}(\mathcal{O}') \to D_{\mathcal{A}}(\mathcal{O})$ as in the statement of the lemma and that moreover this functor commutes with the truncation functors $\tau_{\geq -n}$. We already know that f^* and Rf_* are quasi-inverse equivalence on the corresponding bounded below categories, see Lemma 0D7U. By Lemma 0D6U with $N = 0$ we see that Rf_* indeed defines a functor $Rf_* : D_{\mathcal{A}}(\mathcal{O}) \to D_{\mathcal{A}'}(\mathcal{O}')$ and that moreover this functor commutes with the truncation functors $\tau_{\geq -n}$. Thus for K in $D_{\mathcal{A}}(\mathcal{O})$ the map $f^*Rf_*K \to K$ is an isomorphism as this is true on trunctions. Similarly, for K' in $D_{\mathcal{A}'}(\mathcal{O}')$ the map $K' \to Rf_*f^*K'$ is an isomorphism as this is true on trunctions. This finishes the proof. □